Tuto#4: Multidimensional Design

Exercise 1:

A disposable tableware manufacturing company wishes to set up a decision-making information system in the form of a data mart (a mini data warehouse) to observe its sales activity at the different distribution locations of its items and this in several cities. Their sign, their type (depending on their surface area), their address (postal code and city), their department, their region, indicate these distribution locations. Sales are reported based on a period in months, quarters and years. The number of items according to type, and turnover observes sales.

1- What is the fact to observe for this company?
2- What are the axes of analysis and the measures?
3- Define the star model of this data mart.

Exercise 2:

The Chausséria company wants to build a data warehouse to monitor the evolution of its shoe sales. The Chausséria company has two stores “Chauss_Alger” and “Chauss_Oran” and sells several models of shoes.

1- Propose a conceptual model of the DW_Shoes to observe the evolution of sales in terms of the total number of pairs of shoes sold in relation to the MONTH, STORE and MODEL axes.
2- Propose the corresponding logical model for this data warehouse

Exercice 3

1- Design a star model to analyze the sales of a fast food company. The principle is to measure sales using quantities sold and profits, based on sales made per day, in a given restaurant, for a given food.

The objective is to be able to analyze sales per day, per week, per month and per year. Restaurants can be grouped based on their city and country.

2- Modify this model into a snowflake model to explicitly model the hierarchies of dimensions representing time and geographic location of stores.